Simple Rules, Complex Behavior

Name _____

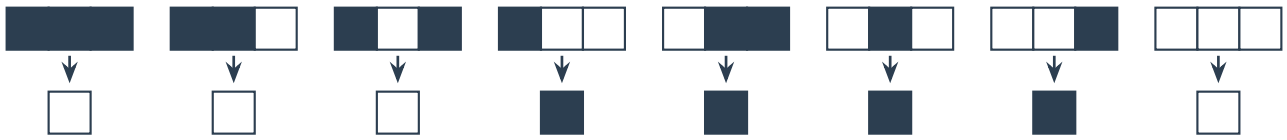
Work with a partner. Compare answers when you finish each problem.

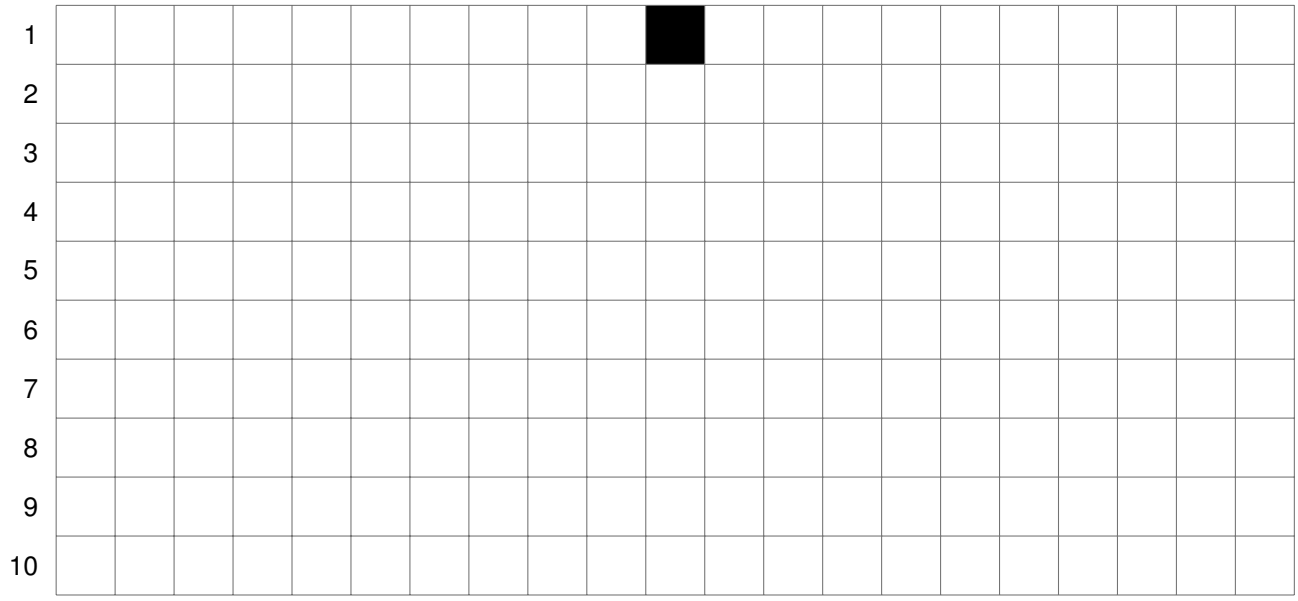
1. The wave. Was anyone in charge of the wave? _____

When many simple parts make something bigger that nobody planned, it is called _____.

2. Be Rule 30. Look at a square and its two neighbors in the row above. Find the same 3 colors on the card. Color the square below. (Outside the grid, everything is white.)

Rule 30: what the middle cell becomes next





Does the middle of your picture have a pattern, or does it look random? _____

3. Explore the rules on screen. Draw each rule. Circle what it looks like.

Rule	What does it look like?			
4	never changes	a neat pattern	chaos	complex
90	never changes	a neat pattern	chaos	complex
30	never changes	a neat pattern	chaos	complex
110	never changes	a neat pattern	chaos	complex

My own rule. I clicked the rule card. My rule number is _____ It looks like _____

4. The Game of Life.

A black cell with _____ black neighbors stays black. A white cell with exactly _____ black neighbors turns black.

What did the glider do? _____

Did any single cell move? _____ So what moved? _____

5. Talk about it. Can simple parts make something smart?

I think simple rules **can** / **cannot** make something smart, because _____

_____

_____